

ORIGINAL ARTICLE

Enhanced thermal stability of (NaCe)-multidoped $\text{CaBi}_2\text{Nb}_2\text{O}_9$ by A-site vacancies-induced pseudo-tetragonal distortionGang Liu  | Shengqiang Ren | Chao Wu | Dan Wang | Fenglian Li |Jiagang Wu  | Qiang Chen

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Abstract

Although A-site codoping of alkali metal and rare-earth elements are known to effectively promote the d_{33} of $\text{CaBi}_2\text{Nb}_2\text{O}_9$ (CBN), its thermal depoling behavior cannot be well promised. In this work, piezoelectricity of CBN was effectively promoted by (NaCe) modification, and its thermal instability was caused by the lattice evolution from orthorhombic structure to pseudo-tetragonal structure during thermal depoling process. A-site vacancies could induce pseudo-tetragonal distortion in $\text{Ca}_{0.92}(\text{Na}_{0.5}\text{Ce}_{0.5})_{0.08}\text{Bi}_2\text{Nb}_2\text{O}_9$ ceramic, which obviously enhanced the thermal stability of lattice by reducing the rotation of the Nb–O octahedron during the thermal depoling process, as compared with orthorhombic structure, thereby weakening the swing of spontaneous polarization (P_s) and promoting the thermal depoling performance. Consequently, good thermal stability was obtained (remained 92% of initial d_{33} after depoling at 600°C). Moreover, dominant role of oxygen vacancies in the electrical conduction and relaxation process promised that the introduction of small amount of A-site vacancies would not influence the electrical properties obviously.

KEYWORDS

aurivillius-type materials, In situ XRD, thermal depoling, XRD rietveld refinement

1 | INTRODUCTION

Lead zirconate titanate (PZT) ceramics, excellent piezoelectric materials, have been widely used in electronic devices (<300°C), like piezoelectric transducers, sensors, and actuators, because of its excellent electrical response around the morphotropic phase boundary (MPB).^{1,2} Taking environmental and human health into consideration, lead-free ceramics, such as $\text{K}_{0.5}\text{Na}_{0.5}\text{NbO}_3$ (KNN), $\text{Na}_{0.5}\text{Bi}_{0.5}\text{TiO}_3$ (BNT), and BaTiO_3 (BT), have been extensively investigated and obtain excellent electric properties via regulating phase boundary.^{3–5} However, when the operating temperature is higher than 450°C, the comparatively low Curie temperature (T_C) of KNN ($T_C \sim 415^\circ\text{C}$) and BT ($T_C \sim 120^\circ\text{C}$), and the depolarization at around 100°C (T_d) for BNT restrict their possible use in high-temperature applications.^{6,7}

The aerospace, aircraft, and nuclear power industries have an indispensable demand for relevant electronic devices, and Aurivillius ferroelectrics have attracted much attention for their brilliant application prospect in high-temperature region, owing to their high Curie temperature (T_C) and large resistivity.^{8–11} The general formula of Aurivillius ferroelectrics is $(\text{Bi}_2\text{O}_2)^{2+}(\text{A}_{m-1}\text{B}_m\text{O}_{3m+1})^{2-}$, where A is a mono-, di-, or trivalent element (or combination) with dodecahedral coordination. B is a transition element suited to octahedral coordination, and m is the number of octahedral layers in the perovskite slab, which can vary from 1 to 6.^{12,13}

$\text{CaBi}_2\text{Nb}_2\text{O}_9$ (CBN), a typical Aurivillius-type compound with $m = 2$, possesses the highest T_C ($\sim 940^\circ\text{C}$) among Aurivillius ferroelectrics.¹⁴ However, it shows a low piezoelectric coefficient (d_{33}) of only 6 pC/N, which is attributed to its high coercive field (E_c) and restricted

spontaneous polarization along *a*-axis.¹⁴ With grain orientation techniques, like template grain growth method (TGG) and spark plasma sintering (SPS), the d_{33} of CBN can be obviously promoted, but the high conductivity along the maximum d_{33} direction (*a*-*b* plane) makes the poling process difficult and limits its integration in devices at high temperature in which conductivity increases.^{14–17} In addition, from the viewpoint of commercial applications, it is desirable to enhance the d_{33} from traditional chemical doping and ordinary firing method. For CBN ceramics, A-site codoping of alkali metal and rare-earth elements can effectively promote the d_{33} , while their thermal depoling behaviors cannot be well guaranteed.^{18–21} However, few researches concern the cause of the unsatisfactory thermal stability, which is important for materials design and practical applications.

Herein, we found that the thermal instability of $\text{Ca}_{0.92}(\text{Na}_{0.5}\text{Ce}_{0.5})_{0.08}\text{Bi}_2\text{Nb}_2\text{O}_9$ was caused by the lattice evolution from orthorhombic structure to pseudo-tetragonal structure during the temperature ascent process. A-site vacancies could induce pseudo-tetragonal distortion in $\text{Ca}_{0.92}(\text{Na}_{0.5}\text{Ce}_{0.5})_{0.08}\text{Bi}_2\text{Nb}_2\text{O}_9$ ceramic, and this distortion reduced the rotation of the Nb–O octahedron during the thermal depoling process, thus weakening the swing of spontaneous polarization (P_s) and promoting the thermal depoling performance. Moreover, dominant role of oxygen vacancies in the electrical conduction and relaxation process guaranteed that the electrical properties of ceramics were almost independent of A-site vacancies.

2 | EXPERIMENTAL PROCEDURE

Samples of compositions $\text{Ca}_{0.92}(\text{Na}_{0.5-x}\square_x\text{Ce}_{0.5})_{0.08}\text{Bi}_2\text{Nb}_2\text{O}_9$ (abbreviated as CBNNC-V100 x , $x = 0$ –0.20 with an interval of 0.05) were prepared by a conventional solid-state reaction method. The weighted raw materials of Bi_2O_3 (99.9%), CaCO_3 (99%), Nb_2O_5 (99.9%), Na_2CO_3 (99.8%), and CeO_2 (99.99%) were mixed by planetary mill for 24 hours. Calcination of the milled powders was carried out at 850°C for 4 hours in air, and then these mixtures were milled again in the same condition. After drying, the powders were mixed thoroughly with a PVA binder solution, and then pressed into disks with a pressure of 10 MPa. Finally, the disks were sintering at 1120°C for 4 hours. For electrical measurement, the sintered ceramics were coated with silver paste to form electrodes on both sides and fired at 650°C for 10 minutes. The samples were poled at 180°C in the silicon oil bath by applying *dc* electric field of 9–10 kV/mm for 10 minutes.

The crystal structures for all samples were characterized using X-ray diffractometry (DX2700; Dandong, China). For Rietveld refinement, the as-sintered CBNNC-V0 and CBNNC-V15 ceramics were granulated into powders and

sieved through 400 meshes, and then the powder diffraction data were collected in the 2θ range 10–120° (X'pert Pro, Panalytical, Netherlands). The surface morphologies were analyzed by scanning electron microscopy (SEM, S-3400 N; Hitachi, Japan). The experimental density of sintered ceramics was determined by the Archimedes method. Temperature dependence of the dielectric properties and impedance spectroscopy were measured using a precision LCR analyzer (HP4980A; Agilent, USA) connecting with a programmable furnace. DC resistivity (ρ) as a function of temperature was performed using an electrometer (Keithley 6517B) in a wide temperature range (200–710°C). The piezoelectric coefficient (d_{33}) was determined using a quasi-static piezo- d_{33} meter (ZJ-3A; Institute of Acoustics, Academia Sinica, China). Thermal depoling experiments were conducted by holding the poled samples for 2 hours at each temperature, cooling to room temperature, measuring the d_{33} value, and repeating the procedures up to 920°C.

3 | RESULTS AND DISCUSSION

Figure 1 shows the XRD patterns indexed by the JCPDS 49-0608 ($\text{CaBi}_2\text{Nb}_2\text{O}_9$) of CBNNC-V100 x ceramics. Single phase of 2 layered Aurivillius type structure with strongest (115) diffraction peak is obtained and no secondary phases are found, meaning that as-prepared CBN samples have formed solid solutions. And the patterns are in consistent with the fact that the most intense reflection of (112 m +1) in BLSFs.¹² The split (200)/(020) peaks of $x = 0$ and $x = 0.05$ are the evidence of orthorhombic structure, and the syncretic (200)/(020) peak in $x = 0.10$ –0.20 is a typical symbol of tetragonal structure. To distinguish it from the tetragonal paraelectric phase ($I4/mmm$) above T_C , we define this structure as pseudo-tetragonal structure ($A2_1am$), which has been referred in our previous work and is caused by the decrease in difference between lattices parameters *a* and *b*.²²

The SEM images of CBNNC-V100 x ceramics are given in Figure 2A–E. Dense structure with flat grains is obtained, attributing to the higher grain growth rate along *a*-*b* plane than *c*-axis.^{23,24} The relative density of ceramics is shown in Figure 2F. All ceramics exhibit high relative density (ρ_{rd}) in the scope of 95.8%–97.5% as compared to the theoretical density (7.05 g/cm³, PDF#49-0608). Moreover, with the increase in vacancies, the mean grain size of samples decreases from 3.99 to 2.28 μm and grain size becomes more uniform, revealing that vacancies would inhibit the grain growth of ceramics during the sintering process. This may be caused by the change in interfacial energy induced by the introduction of ionic vacancies.²⁵ Moreover, pseudo-tetragonal

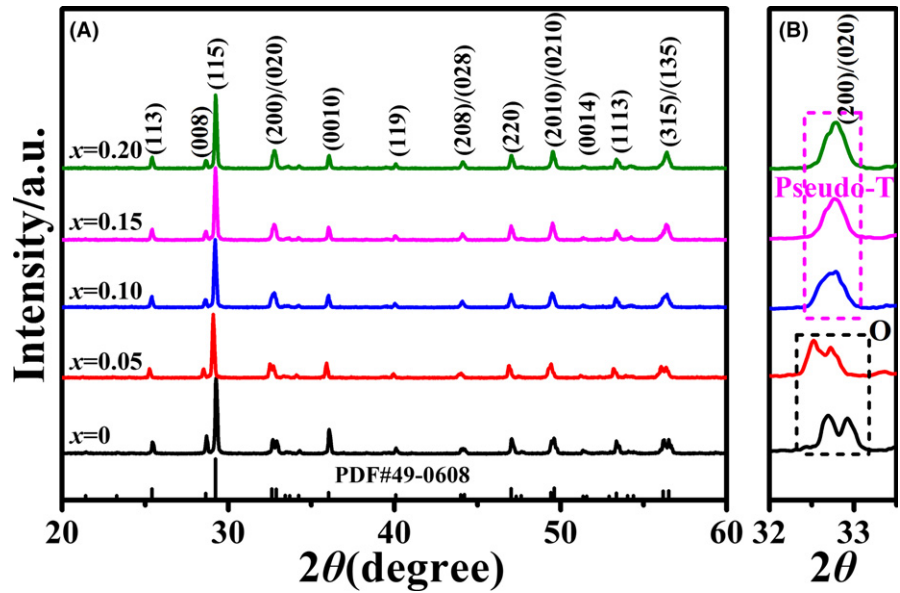


FIGURE 1 (A) XRD patterns and (B) expanded XRD patterns of CBNNC-V100 x ceramics [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

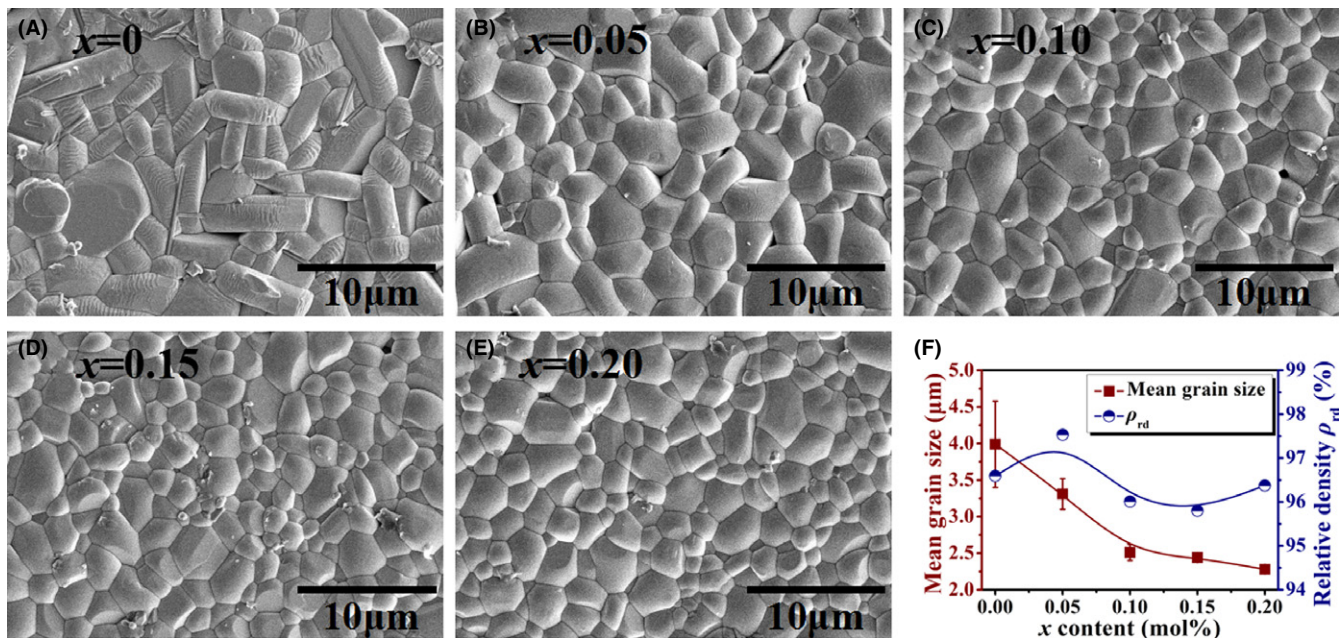


FIGURE 2 (A-E) SEM images of the surfaces and (F) average grain size of CBNNC-V100 x ceramics [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

distortion induced by vacancies may also influence the grain growth process.

Figure 3 exhibits the temperature dependence of the dielectric constant (ϵ_r) and loss tangent ($\tan \delta$) of CBNNC-V100 x ceramics measured at 1 MHz. Single peak near 900°C can be observed, corresponding to Curie temperature (T_C) of phase transition from orthorhombic ferroelectric phase ($A2_1am$) to tetragonal paraelectric phase ($I4/mmm$).^{14,26} With the increase in x , T_C increases about 6–8°C, which can be explained using the tolerance factor t defined as follows:^{21,27}

$$t = (R_A + R_O)/(R_B + R_O) \quad (1)$$

where R_A , R_B , and R_O are ionic radii of A-site cation, B-site cation, and oxygen ion, respectively. Generally, t is used for estimating the structural distortion of perovskite compounds, the closer t is to unity, the closer lattice is to ideal perovskite structure.^{28,29} The ionic radius of Ca^{2+} is about 1.34 Å (12CN),³⁰ and vacancy can be viewed as an ion with small radius. So the introduction of vacancies would decrease the t , correspondingly increase structural distortion, causing the slight promotion of T_C . Moreover,

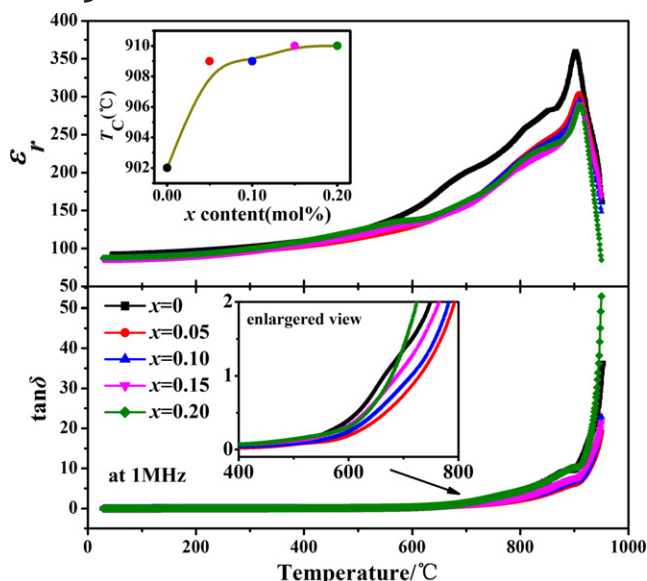


FIGURE 3 Temperature dependence of permittivity (ϵ_r) and loss ($\tan\delta$) of CBNNC-V100 x ceramics [Color figure can be viewed at [wileyonlinelibrary.com](#)]

the introduction of vacancies decrease the dielectric loss in high-temperature region obviously, except for the slight abnormality when $x = 0.20$, which would be beneficial to the better performance of high-temperature devices. Generally, dielectric loss is mainly determined by 2 aspects at high temperature. One is the alternation of polarization state, namely the dielectric constant, another is the electrical conductivity.^{31,32} High dielectric constant and electrical conductivity are adverse to realization of low dielectric loss, vice versa. Introduction of A-site vacancies causes the increase in defects in ceramics, which would restrict the alternation of polarization state, thus leading to the decrease in ϵ_r and decline of $\tan\delta$.²¹ However, the introduced defects would also increase the electrical conductivity, which will be discussed in detail in Figure 10, so $\tan\delta$ rises with the further increase in A-site vacancies.

Piezoelectric constant (d_{33}) and its thermal stability are both key parameters for CBN ceramics, which would determine the performance and reliability of devices used in a wide range of temperature. Figure 4 shows the d_{33} values of CBNNC-V100 x ceramics and their thermal depoling behavior. The introduction of vacancies promotes the d_{33} of ceramics, and CBNNC-V15 possesses the optimal property with d_{33} of 16.1 pC/N. Moreover, the thermal stability of ceramics is also promoted dramatically via introducing vacancy. The samples with pseudo-tetragonal structure ($x = 0.10$ – 0.20) exhibit better thermal depoling performance than orthorhombic structure ($x = 0, 0.05$). When annealing temperature elevates to 600°C, CBNNC-V15 remains 92% of its initial value with d_{33} of 14.8 pC/N, which is much larger than the 79% with 11.9 pC/N of CBNNC-V0.

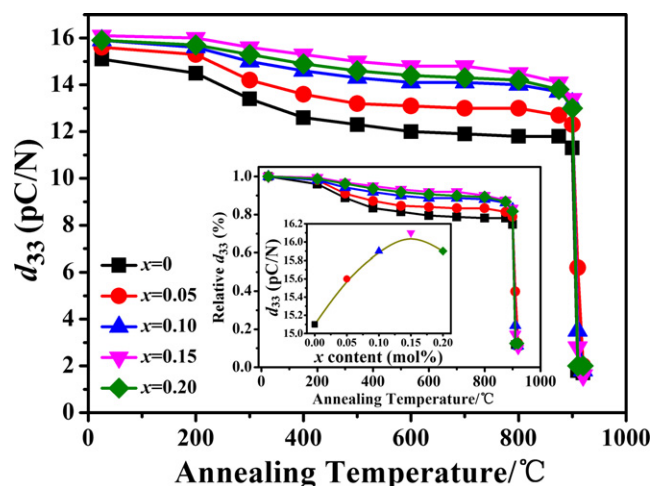


FIGURE 4 Thermal depoling behavior of CBNNC-V100 x ceramics [Color figure can be viewed at [wileyonlinelibrary.com](#)]

In situ X-ray diffraction from room temperature (RT) to 800°C is performed to explore the variation in lattice structure with the elevation of temperature. The XRD patterns of CBNNC-V0 and CBNNC-V15 ceramic powders are given in Figure 5. Both of them show a peak shift to lower angle with the increase in temperature because of the thermal expansion of lattice.³³ The shape and height of (200)/(020) peaks also change with the elevation of temperature. For CBNNC-V0, 2 peaks of (200)/(020) corresponding to orthorhombic structure can be observed from RT to 400°C. However, only one peak of (200)/(020) corresponding to pseudo-tetragonal structure appears when temperature rises up to 600°C. As for CBNNC-V15, only one peak of (200)/(020) corresponding to pseudo-tetragonal structure appears during the whole heating process. Figure 6A and D display the lattice parameters of CBNNC-V0 and CBNNC-V15 at various temperatures, which are obtained by Rietveld refinement with space group $A2_1am$. In CBN ceramics, the lattice parameters of the orthorhombic structure are a , b and c with the relation of $c > a > b$. Higher a/b values give rise to orthorhombic structure and lower a/b gives rise to tetragonal structure. There exists a positive correlation between lattice parameters and temperature, and both lattices are orthorhombic symmetry in the testing temperature range. Furthermore, b increases faster than a with the rise of temperature, which induces the decrease in a/b and gives rise to tetragonal structure. The refinement curves are also given in Figure 6, and the values of reliability factors Sig and R_w are in reasonable range.

Figure 7A and E display the relationship between the thermal stability of d_{33} and the tetragonality of lattice. Tetragonality is defined as a/b and a lower a/b value gives rise to tetragonal structure. The vacancies cause pseudo-tetragonal lattice distortion at room temperature, and the a/b value of CBNNC-V15 is 1.00553, lower than

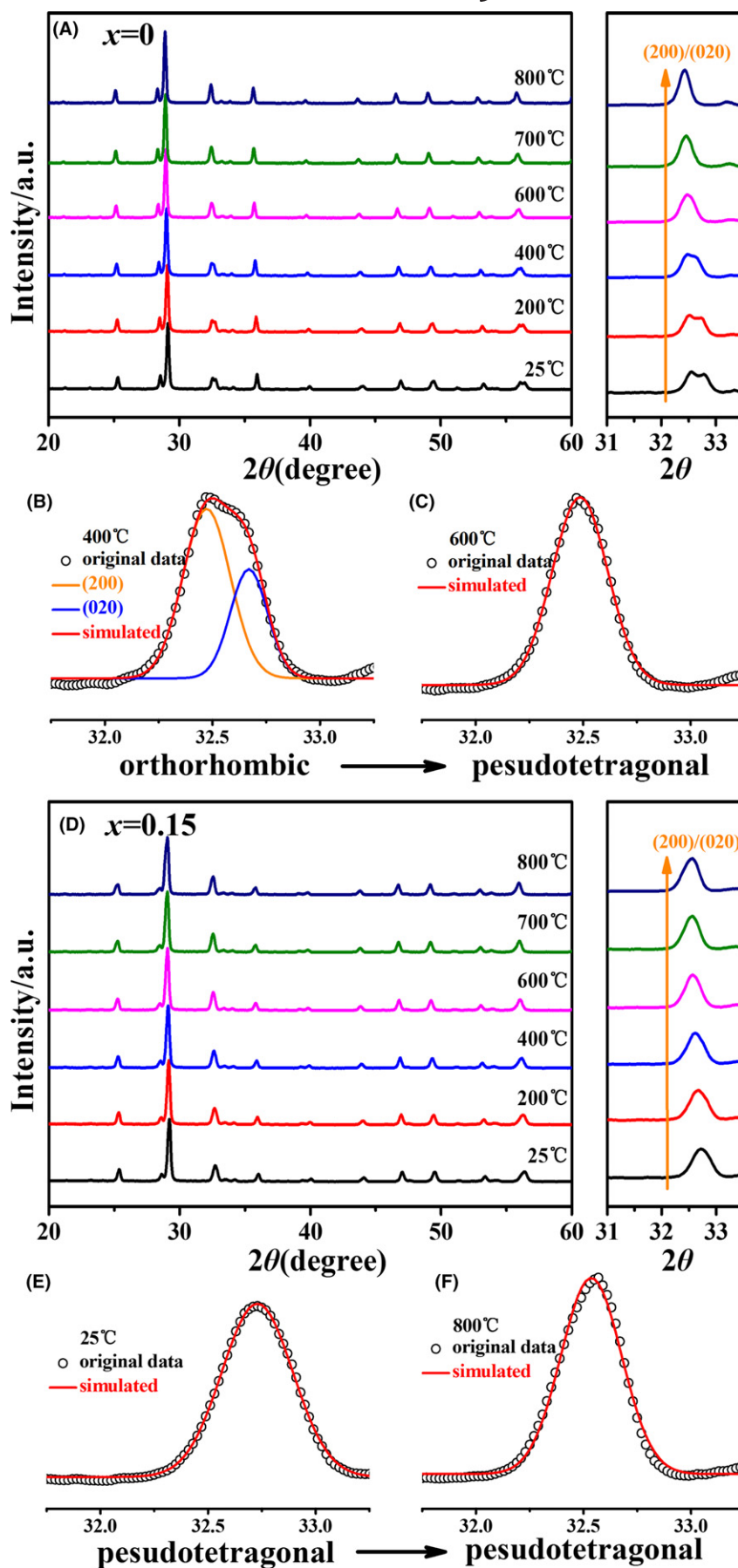


FIGURE 5 (A) In-situ XRD patterns of CBNNC-V0 at various temperature, amplified XRD patterns of CBNNC-V0 ceramics at (B) 400°C and (C) 600°C; (D) In situ XRD patterns of CBNNC-V15 at various temperature, amplified XRD patterns of CBNNC-V15 ceramics at (E) 25°C and (f) 800°C [Color figure can be viewed at wileyonlinelibrary.com]

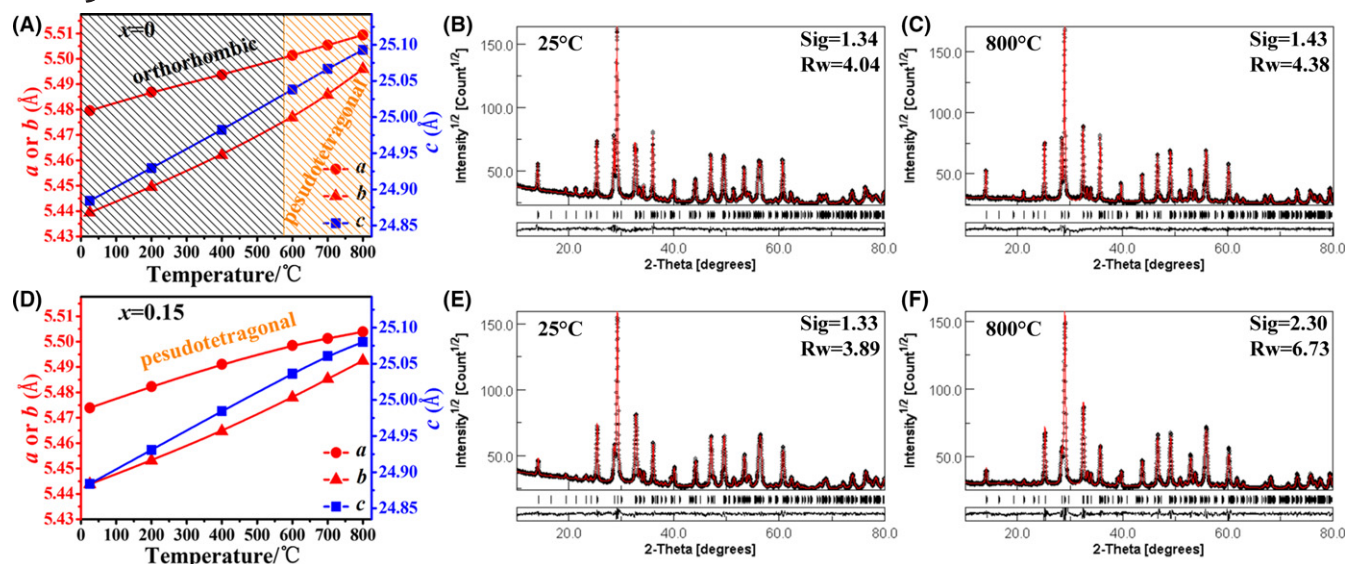


FIGURE 6 (A) Lattice parameters of CBNNC-V0 at various temperature, refinement curves of CBNNC-V0 at (B) 25°C and (C) 800°C; (D) Lattice parameters of CBNNC-V0 at various temperature, refinement curves of CBNNC-V15 at (E) 25°C and (F) 800°C [Color figure can be viewed at wileyonlinelibrary.com]

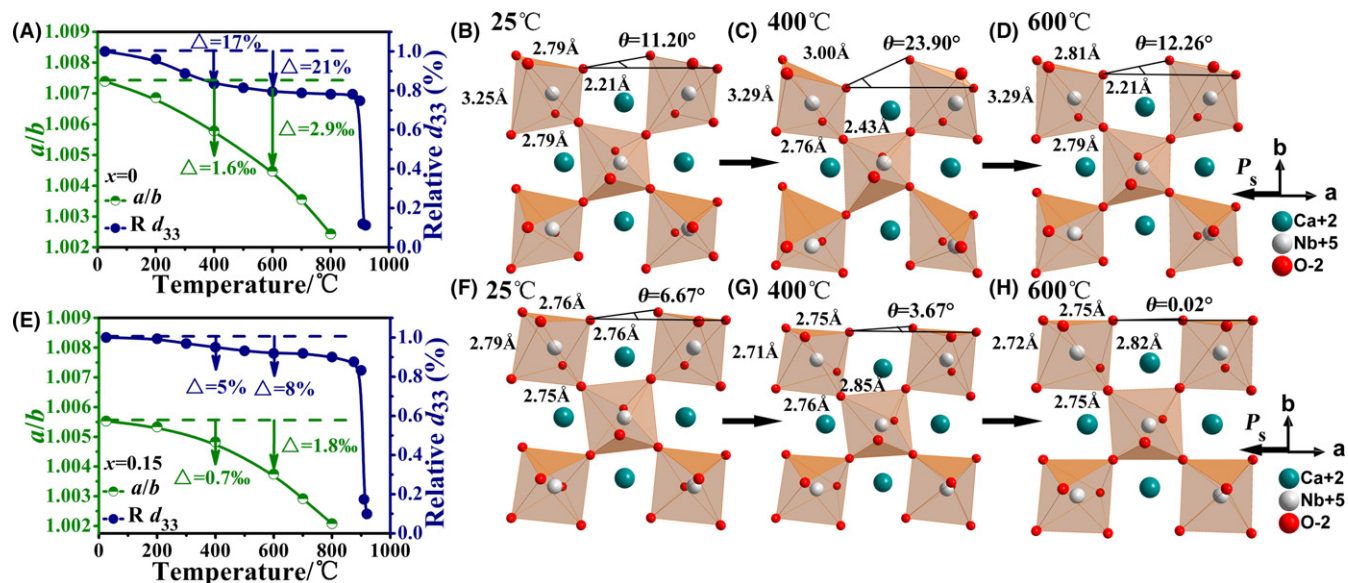


FIGURE 7 (A) Correlations between the thermal stability of d_{33} and the tetragonality of lattice of CBNNC-V0, visualized structure of CBNNC-V0 viewed from <001> direction at (B) 25°C, (C) 400°C and (D) 600°C; (E) Correlations between the thermal stability of d_{33} and the tetragonality of lattice of CBNNC-V15, visualized structure of CBNNC-V15 viewed from <001> direction at (F) 25°C, (G) 400°C and (H) 600°C [Color figure can be viewed at wileyonlinelibrary.com]

the 1.00739 of CBNNC-V0. During the temperature rise period, a/b value decreases with the temperature, but the decline rate are different between them. When temperature rise up to 400 and 600°C, the a/b value of CBNNC-V15 shows a decline of 0.7‰ and 1.8‰, respectively, much lower than the 1.6‰ and 2.9‰ of CBNNC-V0. Correspondingly, the d_{33} value of CBNNC-V15 exhibits a decline of 5% and 8% at 400 and 600°C, much better than the 17% and 21% of CBNNC-V0. CBNNC-V15 with the

pseudo-tetragonal lattice distortion possesses a more stable a/b value, and exhibits a higher thermal stability of d_{33} . Visualized structure of CBNNC-V0 and CBNNC-V15 at the representative temperatures viewed from the <001> direction are also given in Figure 7 according to the atomic coordinates obtained from Rietveld refinement. Due to the existence of glide and mirror planes along the b -axis and c -axis, the spontaneous polarization (P_s) of CBN mainly originates from the displacement along a -axis from the

corresponding positions in the parent tetragonal structure.^{12,14} Moreover, the contribution to P_s mainly originates from the perovskite slab, especially the contribution from Nb–O octahedron.²⁷ The rotation angle labeled as θ in Figure 7, which is estimated from the coordinates of O atoms, is an important parameter to estimate the distortion degree of Nb–O octahedron.³⁴ Comparing with CBNNC-V15, CBNNC-V0 possesses a more unstable Nb–O octahedron during the heating process, in term of the variation in shape and rotation angle of Nb–O octahedron. The unstable octahedron of CBNNC-V0 would cause the swing of P_s , which is adverse to the thermal depoling. For CBNNC-V15, it possesses a relative stable Nb–O octahedron, thus owning a more stable orientation of P_s and exhibiting a better thermal depoling behavior.

Figure 8A and E show the total polarization of CBNNC-V0 and CBNNC-V15 at various temperature, which is calculated using Shimakawa's mode²⁷:

$$P_s = \sum_i \frac{m_i \Delta x_i Q_i e}{V} \quad (2)$$

where m_i is the site multiplicity, Δx_i is the atomic displacement along a -axis from the corresponding position in the tetragonal structure, $Q_i e$ is the ionic charge of the i th

constitution, and V is the volume of the unit cell. Simultaneously, the corresponding patterns of lattice structure are also given. The total P_s remains stable at the representative temperature, even exhibits a slightly increase in CBNNC-V15 when temperature rises. During the temperature rise process, the incline angle labeled as θ of Nb–O octahedron in CBNNC-V0, which is estimated from the coordinates of O(1) and O(2), exhibits small variation, conversely, it shows a striking change in CBNNC-V15. Thus, as for CBNNC-V0, it endures a lattice variation from orthorhombic structure to pseudo-tetragonal structure, resulting in a larger rotation and smaller incline of Nb–O octahedron. While CBNNC-V15 displays a smaller rotation and larger incline of Nb–O octahedron to maintain its pseudo-tetragonal structure.

In the previous work, decreasing grain size actually decreases thermal stability of CBN ceramics due to increase in internal stresses resulting from increased incompatibility at grain boundaries, leading to easier thermal depolarization.³⁵ However, only when the large decrease in grain size happens ($\approx 0.45 \mu\text{m}$ in the previous work),³⁵ it would play a dominant role in the thermal depolarization process. Herein, although grain size decreases with increase in A-site vacancy concentration, the decreasing amplitude

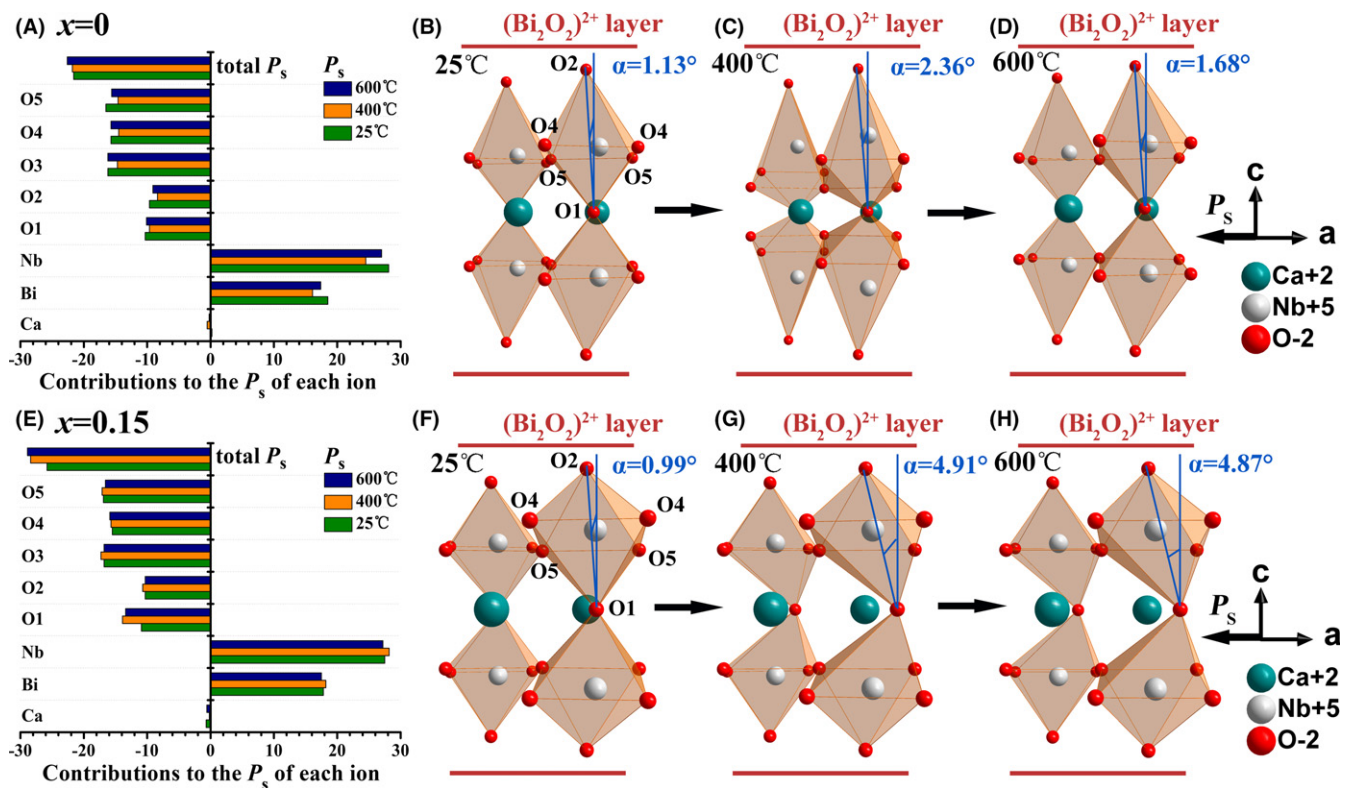


FIGURE 8 (A) Spontaneous polarization (P_s) of CBNNC-V0 at different temperature, visualized structure of CBNNC-V0 viewed from $\langle 010 \rangle$ direction at (B) 25°C, (C) 400°C and (D) 600°C; (E) Spontaneous polarization (P_s) of CBNNC-V0 at different temperature, visualized structure of CBNNC-V0 viewed from $\langle 010 \rangle$ direction at (F) 25°C, (G) 400°C and (H) 600°C [Color figure can be viewed at wileyonlinelibrary.com]

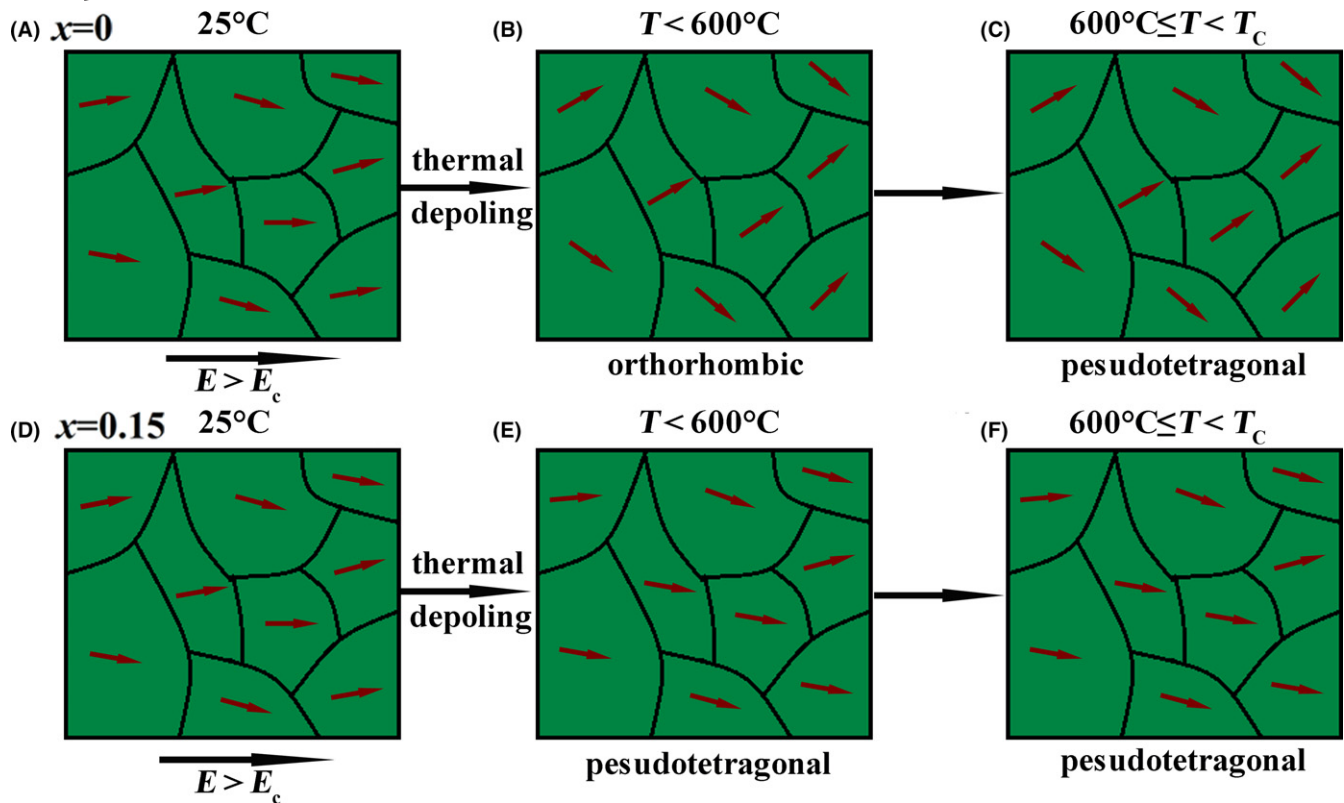


FIGURE 9 The schematic of thermal depoling in (A, B, C) CBNNC-V0 and that of (D, E, F) CBNNC-V15 [Color figure can be viewed at wileyonlinelibrary.com]

is not tremendous, so it would not influence the thermal depolarization greatly. Instead, pseudo-tetragonal distortion induced by A-site vacancies concentration plays a more dominant role in the thermal depolarization process. So in the next discussion, the effect of grain size on thermal stability is not considered. Combining the analysis in Figures 7 and 8, a plausible model of thermal annealing process is proposed. During the polarization process, the random oriented P_s would arrange along the electric field direction when $E > E_c$, and the P_s remains an ordered arrangement after removing the poling electric field, as

depicted in Figure 9A and D. Piezoelectric constant (d_{33}) is determined by the component of P_s along the electric field direction, and a larger component would contribute to a higher d_{33} .³⁶ A simple formula is proposed to describe this correlation:

$$\vec{P}_{\vec{E}} = \sum_i \vec{P}_i \cos \langle \vec{P}_i, \vec{E} \rangle \quad (3)$$

where $\vec{P}_{\vec{E}}$ is the total component of P_s along the electric field direction, $\vec{P}_i$ is the polarization of the i th grain and $\langle \vec{P}_i, \vec{E} \rangle$ is the included angle between $\vec{P}_i$ and $\vec{E}$. During

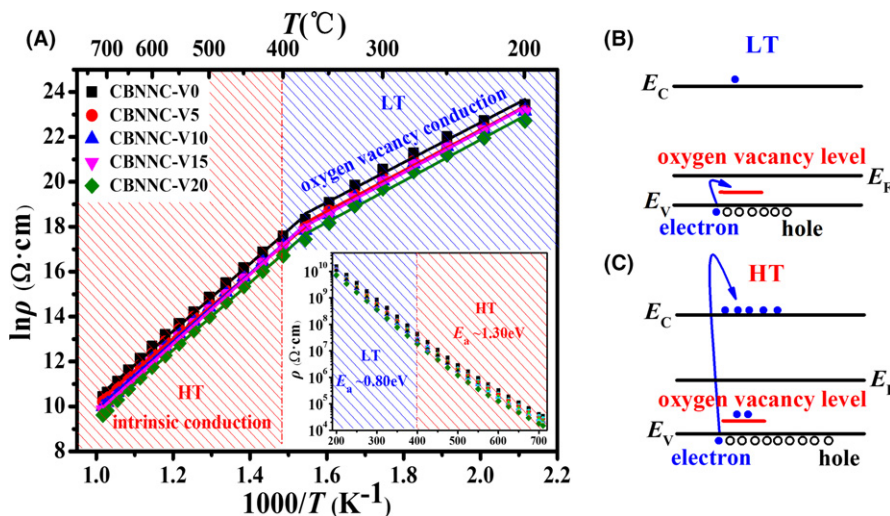


FIGURE 10 (A) The temperature dependence of dc resistivity of CBNNC-V100x ceramics, approximate band structure of CBNNC-V100x at (B) low-temperature region (LT) and (C) high-temperature region (HT) [Color figure can be viewed at wileyonlinelibrary.com]

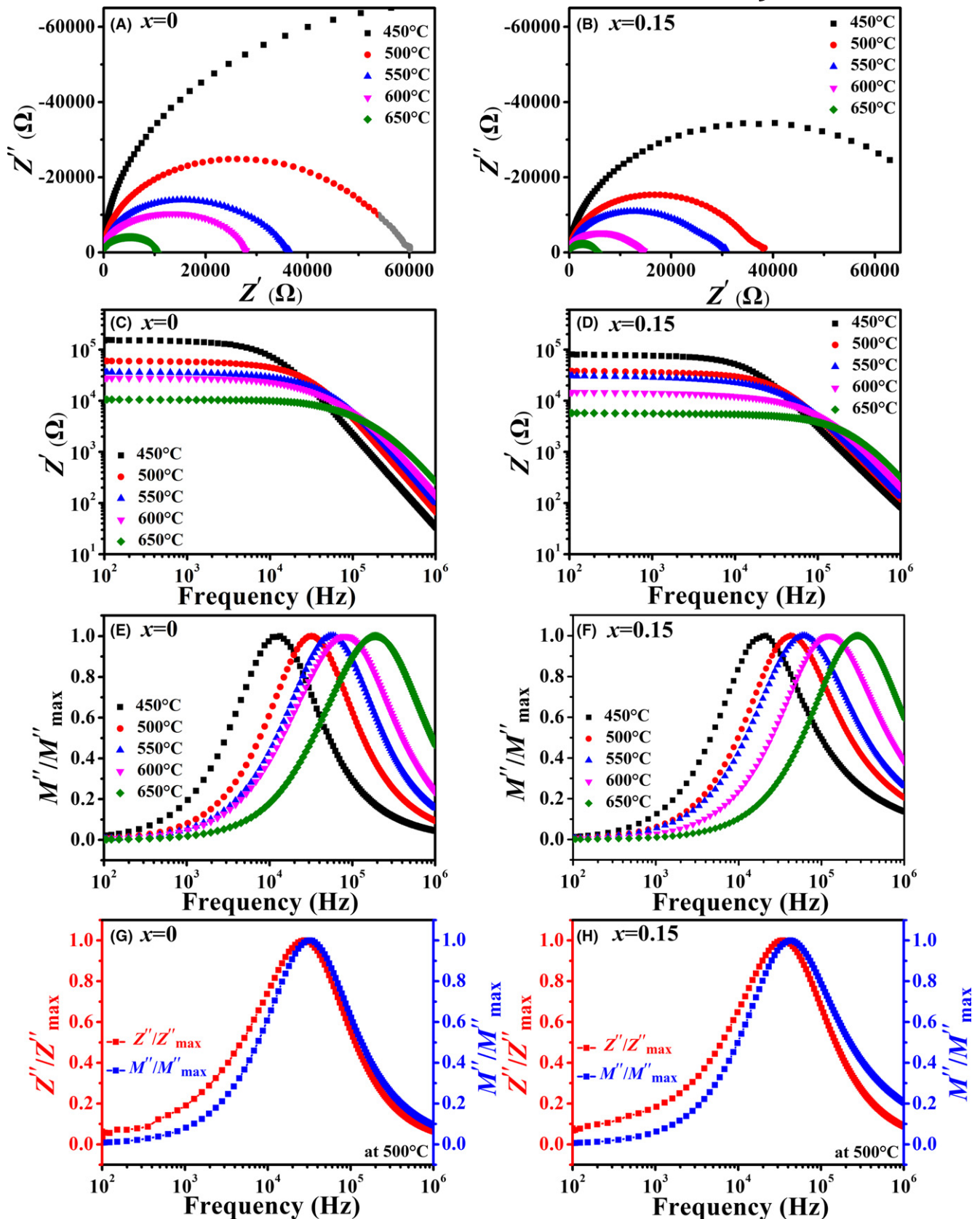


FIGURE 11 (A, B) Cole-Cole plots of CBNNC-V0 and CBNNC-V15 at different temperature, (C, D) frequency dependence of real part (Z') of CBNNC-V0 and CBNNC-V15 at different temperature, (E, F) normalized imaginary part of modulus $M''/M''_{\max}$ at various temperature of CBNNC-V0 and CBNNC-V15 as a function of frequency, (G, H) $Z''/Z''_{\max}$ and $M''/M''_{\max}$ of CBNNC-V0 and CBNNC-V15 as a function of frequency at 500°C [Color figure can be viewed at wileyonlinelibrary.com]

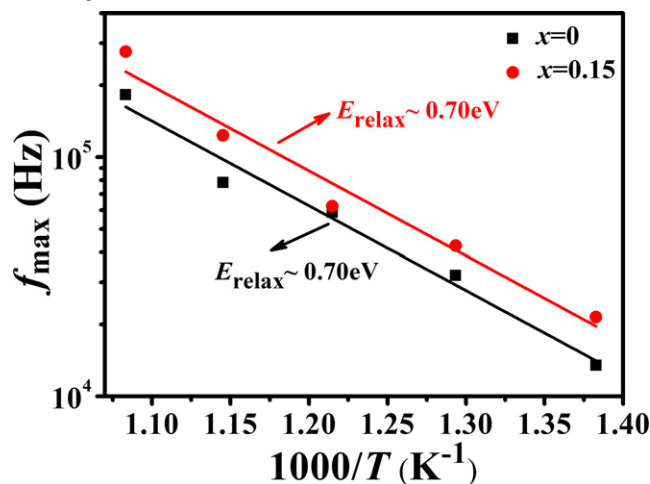


FIGURE 12 Theoretical fitting of relaxation frequency of CBNNC-V0 and CBNNC-V15 [Color figure can be viewed at wileyonlinelibrary.com]

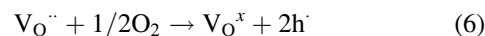
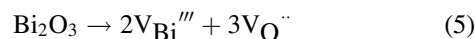
the thermal depoling process, the $\vec{P}_i \cdot \vec{E}$ decrease with the rise of temperature, which would leads to the decline of d_{33} . Decrease in $\vec{P}_i$ or increase in $\langle \vec{P}_i, \vec{E} \rangle$, or both can cause the reduction in $\vec{P}_i \cdot \vec{E}$. However, as mentioned above, P_s of CBNNC-V0 and CBNNC-V15 are nearly independent of temperature, so $\vec{P}_i$ remains stable and the thermal stability of $\vec{P}_i \cdot \vec{E}$, namely d_{33} , is determined by the thermal stability of $\langle \vec{P}_i, \vec{E} \rangle$. For CBNNC-V0, as Figure 9B shows, $\langle \vec{P}_i, \vec{E} \rangle$ would be unstable when annealing temperature is lower than 600°C, which is attributed to the unstable Nb–O octahedron caused by the lattice evolution from orthorhombic structure to pseudo-tetragonal structure during thermal depoling process, thus causing the thermal instability of d_{33} . When temperature is between 600°C and T_C , as Figure 9C shows, stable $\langle \vec{P}_i, \vec{E} \rangle$ corresponding to the pseudotetragonal structure contributes to the more stable thermal stability of d_{33} in CBNNC-V0. On the contrary, the stable Nb–O octahedron corresponding to the pseudo-tetragonal structure of CBNNC-V15 would contribute to a much more stable $\langle \vec{P}_i, \vec{E} \rangle$ and higher thermal stability of d_{33} until T_C , as shown in Figure 9E and F.

Figure 10 shows the dc resistivity (ρ) as a function of temperature that can be fitted by the Arrhenius equation³⁷:

$$\rho = \rho_0 \exp(E_a/kT) \quad (4)$$

where the E_a is the activation energy of the charge carriers, k is the Boltzmann constant and T is the absolute temperature. During the sintering process, the volatilization of Bi is a general concern for BLSF material. It would generate oxygen vacancies due to the charge neutrality,^{26,38} thus

resulting in the p -type conduction mechanism of ceramics:



Na-deficiency would further increase the amount of oxygen vacancies:



so Na-deficiency can be viewed as acceptor dopant and the resistivity of ceramics decrease gradually with the increase in x . However, the decreasing amplitude is small. The fitted E_a of all samples show no regular variation with respect to the x , and is in the range 0.77–0.80 eV. As we know, E_a can reflect the approximate position of Fermi level in the band gap with the formula of $E_a \approx E_F - E_V$ in the p -type semiconductor,^{39–41} where the E_F is the position of Fermi level and E_V is the valence-band edge energy, respectively. The more acceptor dopant is, the closer is E_F to E_V . The introduction of vacancies do not obviously change the position of E_F , meaning that the conduction at low-temperature region (LT) dominated by oxygen vacancies with electron hopping from valance band to acceptor level, as shown in Figure 10B, so a small amount of introduced vacancies will not shift this impurity energy level. At high-temperature region, E_a is in the range 1.25–1.33 eV, corresponding to intrinsic conduction with electron hopping from valance band to conduction, as shown in Figure 10C. Generally, the intrinsic E_a value could be close to the half energy of the band gap E_g ,^{21,42} thus the approximate equivalent E_a implies that vacancy will not influence the band gap of ceramics obviously.

Figure 11A and B present the Cole-Cole plots of CBNNC-V0 and CBNNC-V15 ceramics. Only a single semicircle is observed for the 2 samples at the testing temperature range, indicating a dominant behavior of grain response.⁴³ The small “tail” of CBNNC-V15 appears at low-frequency range may be attributed to the enhanced grain-boundary effect caused by the increased grain-boundary area. Both semicircles show a decrease in diameter with the elevation of temperature, manifesting the negative temperature coefficient of resistance. Figure 11C and D displays the real part (Z') of complex impedance (Z^*) as a function of frequency at different temperature. The frequency-independent plateau at low frequency and a frequency-dependent region at high frequency correspond to direct current (dc) dominant region and alternating current (ac) dominant region, respectively.⁴⁴ Comparing with CBNNC-V0, CBNNC-V15 expresses a decrease in Z' value accompanied by the shift of frequency-dependent plateau to higher frequency at the same testing temperature. This phenomenon can be attributed to the enhanced dc

conductive effect caused by the increased charge carriers. Figure 11E and F show the normalized imaginary part ($M''/M''_{\max}$) of complex modulus (M^*) vs frequency at different temperature. The frequency region below the peak maximum $M''_{\max}$ determines the long-range motion of ions contributing to *dc* conductivity. At high frequency above the peak maximum, carriers are confined in potential wells and move in short distance, contributing to *ac* conductivity.⁴⁵ Only one peak corresponding to the grain is displayed because of its much smaller capacitance value than that of grain boundary. The shift of $M''_{\max}$ to higher frequency with the increase in temperature implies the temperature-dependent relaxation process, and the conductivity process and electric polarization are thermally activated.⁴⁶ Figure 11G and H show the normalized imaginary part M'' and Z'' as functions of frequency at 500°C. As we know, the overlap of $Z''_{\max}$ and $M''_{\max}$ represents the long-range conduction, while the separation between $Z''_{\max}$ and $M''_{\max}$ is the evidence of the localized conduction and a departure from an ideal Debye-like behavior where the $Z''_{\max}$ and $M''_{\max}$ are frequency coincident.⁴³ The more obvious departure $Z''_{\max}$ and $M''_{\max}$ of CBNNC-V15 than CBNNC-V0 further indicate that the oxygen vacancies dominate the conduction and relaxation process.

Figure 12 depicts the peak frequency corresponding to the $M''_{\max}$ as a function of testing temperature. The activation energy E_{relax} involved in the relaxation process of ions can be obtained from the Arrhenius equation:

$$f_{\max} = f_0 \exp(-E_{\text{relax}}/kT) \quad (8)$$

while $f_{\max}$ is the frequency corresponding to the $M''_{\max}$, f_0 is the ion hopping attempt frequency, and E_{relax} is the activation energy of dielectric relaxation process. The E_a of CBNNC-V0 and CBNNC-V15 are both near 0.70 eV, indicating the increased oxygen vacancies would not obviously influence the relaxation process. The value f_0 of CBNNC-V0 and CBNNC-V15 are 1.12×10^9 and 1.60×10^9 Hz, respectively, and the higher value of CBNNC-V15 may be attributed to the increased oxygen vacancies sites for ion hopping.⁴⁷

4 | CONCLUSIONS

Thermal instability of $\text{Ca}_{0.92}(\text{Na}_{0.5}\text{Ce}_{0.5})_{0.08}\text{Bi}_2\text{Nb}_2\text{O}_9$ ceramic was caused by the lattice evolution from orthorhombic structure to pseudo-tetragonal structure during thermal depoling process. The introduction of A-site vacancies-induced pseudo-tetragonal distortion in $\text{Ca}_{0.92}(\text{Na}_{0.5}\text{Ce}_{0.5})_{0.08}\text{Bi}_2\text{Nb}_2\text{O}_9$ ceramic, which enhanced its thermal stability via reducing the rotation of the Nb–O octahedron during the thermal depoling process, thereby weakening the

swing of spontaneous polarization (P_s) and promoting the thermal depoling performance. Consequently, CBNNC-V15 with pseudo-tetragonal structure remained 92% of initial d_{33} after depoling at 600°C, much better than 79% of CBNNC-V0 with orthorhombic structure. Moreover, the dominant role of oxygen vacancies in electrical conduction and relaxation process ensured that the introduction of small amount of A-site vacancies would not influence the electrical properties obviously. This work may provide a new viewpoint to achieve the high piezoelectric constant and well thermal stability in CBN via regulating lattice structure.

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